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RBE211TC DYNAMIC SYSTEMS

LAGRANGIAN METHOD

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Lecture Outline

1. Energy-based modelling
2. Generalized coordinates
3. Lagrange’s equation
4. Procedure for deriving dynamic equations
5. Worked examples

Why Do We Need Another Modelling Method?

- The Lagrangian method is especially useful for:
  - systems with **multiple degrees of freedom**
  - systems with **translational and rotational motion**
  - systems with **constraints**
  - robotic systems with multiple joints and coupled motion
  - systems where energy expressions are easier to formulate than direct balance equations

- **Key idea:**
  - Use energy relationships to derive the governing dynamic equations.

Generalized Coordinates

- A **generalized coordinate** is a variable that uniquely describes the configuration of a system.

- Examples:
  – translational displacement: $\boldsymbol{x}$
  – angular displacement: $\boldsymbol{\theta}$

- Depending on the system, a generalized coordinate may represent displacement, rotation, charge, or another configuration variable.

Lagrange's Equation

- For each generalized coordinate q_i,

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i$$

where

T: kinetic energy

V: potential energy

D: dissipation function

Q_i: generalized force/input

Procedure for Applying Lagrangian Method

1. Choose the generalized coordinate(s) q_i
2. Write the kinetic energy T
3. Write the potential energy V
4. Write the dissipation function D, if damping is present
5. Identify the generalized input Q_i
6. Substitute into Lagrange's equation
7. Simplify to obtain the governing dynamic equation(s)

Useful Energy Expressions

- Kinetic energy:

$$T_{trans} = \frac{1}{2} m\dot{x}^2$$

$$T_{rot} = \frac{1}{2} J\dot{\theta}^2$$

- Potential energy:

$$V_{spring} = \frac{1}{2} kx^2$$

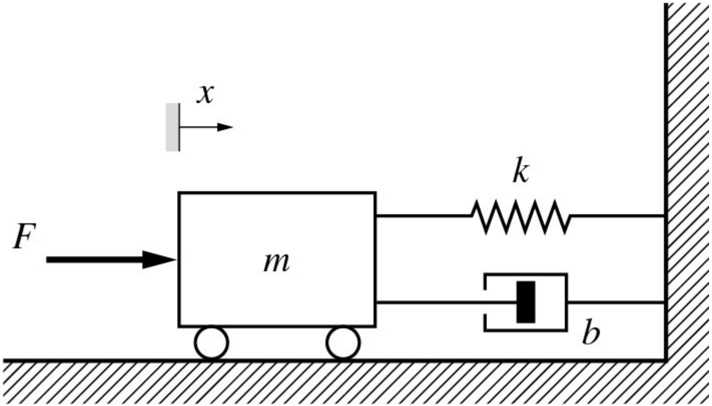
$$V_{grav} = mgh$$

- Dissipation function:

$$D_{viscous} = \frac{1}{2} b\dot{x}^2$$

Example 1

Use Lagrange’s equation to derive the governing equation of motion for the system shown.



Generalized coordinate: $q = x$

Example 1

The generalized coordinate is

$$q = x$$

The kinetic energy of the system is given by

$$T = \frac{1}{2}m\dot{x}^2$$

Thus

Example 1

The potential energy of the system is given by

$$V = \frac{1}{2}kx^2$$

Thus

The dissipation function is given by

$$D = \frac{1}{2}b\dot{x}^2$$

Thus

Example 1

The generalized input associated with the coordinate x is

$$Q = F(t)$$

The equation of motion is obtained by substituting the obtained expressions into Lagrange's equation

and is found to be given by

$$\boxed{m\ddot{x} + b\dot{x} + kx = F(t)}$$

Lagrange's Equation

Example 1

Derived using Lagrange's method:

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

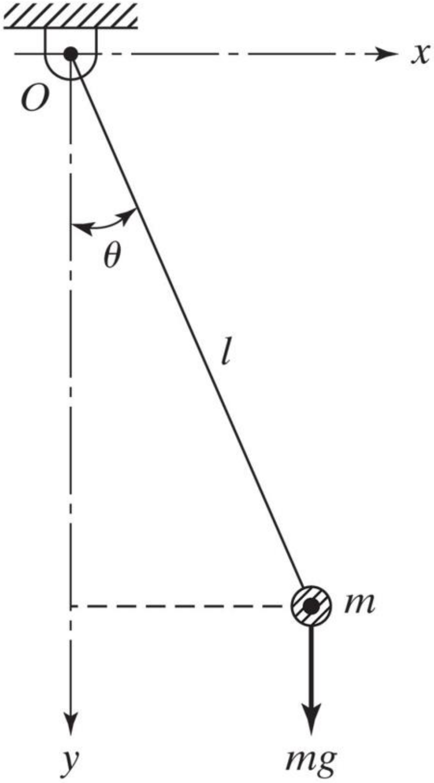
This is the same governing equation obtained using Newton's second law.

Observation:

Different modelling route, same physical model.

Example 2

Use Lagrange’s equation to derive the governing equation of motion for the system shown.



Generalized coordinate: $q = \theta$

Lagrange's Equation

Example 2

The generalized coordinate is

$$q = \theta$$

The kinetic energy of the system is given by

$$T = \frac{1}{2}m\left(l\dot{\theta}\right)^2 = \frac{1}{2}ml^2\dot{\theta}^2$$

Thus

Example 2

The potential energy of the system is given by

$$V = mgl(1 - \cos\theta)$$

Thus

Assume the damping is negligible and there is no external applied torque, so

Example 2

The equation of motion is obtained by substituting the obtained expressions into Lagrange's equation

and is found to be given by

$$\boxed{ml^2\ddot{\theta} + mgl\sin\theta = 0}$$

Example 2

Pendulum equation:

$$ml^2\ddot{\theta} + mgl\sin\theta = 0$$

For small angular displacement:

$$\sin\theta \approx \theta$$

Hence,

$$ml^2\ddot{\theta} + mgl\theta = 0$$

or

$$\boxed{\ddot{\theta} + \frac{g}{l}\theta = 0}$$

Example 2

Using the Lagrangian method, we obtained:

- a **nonlinear** pendulum model:

$$ml^2\ddot{\theta} + mgl\sin\theta = 0$$

- and, for small angles, a **linearized** model:

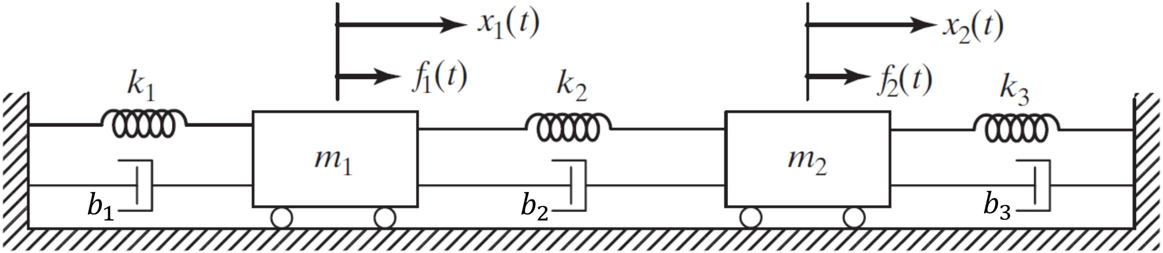
$$\ddot{\theta} + \frac{g}{l}\theta = 0$$

Observation:

The same modelling method can naturally handle both nonlinear and linearized system descriptions.

Example 3

Use Lagrange's equation to derive the governing equations of motion for the system shown.



Generalized coordinates: $q_1 = x_1, q_2 = x_2$

Example 3

Let $q_1 = x_1$, $q_2 = x_2$. The total kinetic energy of the system is

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2$$

Thus,

$$\frac{\partial T}{\partial \dot{x}_1} =$$

$$\frac{\partial T}{\partial \dot{x}_2} =$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}_1}\right) =$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}_2}\right) =$$

$$\frac{\partial T}{\partial x_1} =$$

$$\frac{\partial T}{\partial x_2} =$$

Example 3

The total potential energy of the system is

$$V = \frac{1}{2}k_1 x_1^2 + \frac{1}{2}k_2(x_1 - x_2)^2 + \frac{1}{2}k_3 x_2^2$$

Thus,

$$\frac{\partial V}{\partial x_1} = \qquad\qquad \frac{\partial V}{\partial x_2} =$$

Example 3

The dissipation function of the system is

$$D = \frac{1}{2} b_1 \dot{x}_1^2 + \frac{1}{2} b_2 (\dot{x}_1 - \dot{x}_2)^2 + \frac{1}{2} b_3 \dot{x}_2^2$$

Thus,

$$\frac{\partial D}{\partial \dot{x}_1} =$$

$$\frac{\partial D}{\partial \dot{x}_2} =$$

Example 3

The generalized external inputs associated with the coordinates are

$Q_1 =$ $Q_2 =$

Substitute into Lagrange's equation gives

x_1:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}_1}\right) - \frac{\partial T}{\partial x_1} + \frac{\partial V}{\partial x_1} + \frac{\partial D}{\partial \dot{x}_1} = f_1$$

$$m_1\ddot{x}_1 + (b_1 + b_2)\dot{x}_1 - b_2\dot{x}_2 + (k_1 + k_2)x_1 - k_2x_2 = f_1$$

x_2:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}_2}\right) - \frac{\partial T}{\partial x_2} + \frac{\partial V}{\partial x_2} + \frac{\partial D}{\partial \dot{x}_2} = f_2$$

$$m_2\ddot{x}_2 - b_2\dot{x}_1 + (b_2 + b_3)\dot{x}_2 - k_2x_1 + (k_2 + k_3)x_2 = f_2$$

Lagrange's Equation

Example 3

Therefore, the governing equations of motion of the system are

$$x_1: \quad \boxed{\begin{aligned} &m_1\ddot{x}_1 + (b_1 + b_2)\dot{x}_1 - b_2\dot{x}_2 + (k_1 + k_2)x_1 - k_2x_2 = f_1 \\ &m_2\ddot{x}_2 - b_2\dot{x}_1 + (b_2 + b_3)\dot{x}_2 - k_2x_1 + (k_2 + k_3)x_2 = f_2 \end{aligned}} \quad :x_2$$

In matrix form:

Review

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